DAY 2 | EXCEL

My Data Analyst Journey

Formulas, Functions & Conditional Formatting

Day 2 of learning Excel in public.

= Alt Ctrl F4

Swipe to learn with me >

Formula Basics

Formula vs Function

Formula vs Function

Formula = you write it manually. Function = a predefined calculation built into Excel.

The = Sign

Every formula starts with =. It tells Excel to calculate.

How It Works

Type = in A1, then click any cell (like B2). Its address appears in A1.

Built From

Numbers, text, characters and cell references.

=100+200

Hardcoded. It never changes.

=A1+B1

Dynamic. It updates when A1 or B1 changes.

Operators

The 4 basics, plus 2 extras

`=A1+B1`	**Add**
`=A1-B1`	**Subtract**
`=A1*B1`	**Multiply**
`=A1/B1`	**Divide**
`=A1^2`	**Power**
`=A1&B1`	**Join text**

> *Always use cell references (A1) instead of typed numbers so results update automatically.*

SUM & AutoSum

Add numbers the smart way

`=SUM(A2,B2)` — **Separate cells**

`=SUM(A2:C2)` — **Range: A2 + B2 + C2**

`=SUM(A1:A3)` — **Continuous range**

`=SUM(A1,A3)` — **Only A1 and A3 (skips A2)**

AutoSum: click the cell below your numbers, press Alt + =, then Enter. Drag the Fill Handle (bottom-right corner) to copy a formula down - references adjust automatically.

Core Functions

Type the name, press Tab

`=PRODUCT(A2:C2)`	Multiply a range
`=AVERAGE(G3:H3)`	Average (mean)
`=MAX(A2:C2)`	Largest value
`=MIN(A2:C2)`	Smallest value
`=ROUND(A1,2)`	Round to 2 decimals
`=TODAY()`	Today's date

*=A2*B2 and =PRODUCT(A2:B2) give the same answer. Use PRODUCT when you have a range.*

COUNT Functions

Row example: ab | 85 | 86 | (blank)

`=COUNT(C2:F2)` **Numbers only = 2**

`=COUNTA(C2:F2)` **Non-empty cells = 3**

`=COUNTBLANK(C2:F2)` **Empty cells = 1**

COUNT ignores text and blanks. COUNTA counts anything that is filled. COUNTBLANK counts only empty cells.

DAY 2 | EXCEL

Day 2 Shortcuts

Work faster with formulas and data

AutoSum	Alt + =
Select multiple cells	Ctrl + Click
Fill all selected cells	Ctrl + Enter
Go To / Go To Special	Ctrl + G
Find & Replace	Ctrl + H
Find	Ctrl + F
Lock a cell (A1)	F4
Undo (Ctrl Y = Redo)	Ctrl + Z

Fill blanks fast: Ctrl+G > Special > Blanks, type 0, then Ctrl+Enter.

Paste Special

Ctrl + Alt + V, then press a letter

Values only (the result)	Ctrl + Alt + V + V
Formulas only	Ctrl + Alt + V + F
Formats only (color, border)	Ctrl + Alt + V + T
All (data, formulas, formats)	Ctrl + Alt + V + A
Transpose (rows <-> columns)	Ctrl + Alt + V + E
Multiply destination cells	Ctrl + Alt + V + M
Column widths	Ctrl + Alt + V + W

After the letter, press Enter. Ctrl+C copies, Ctrl+X cuts, Ctrl+V pastes everything.

Conditional Formatting

Highlight cells by condition

What It Does

Colors cells automatically when a condition is true. Example: sales above 50,000 turn green.

Where

Home tab > Styles > Conditional Formatting.

Steps

1) Select the column (Ctrl+Shift+Down). 2) Conditional Formatting > Highlight Cells Rules. 3) Pick a condition, value and color > OK.

The Menu

Highlight Cells Rules, Top/Bottom Rules, Data Bars, Color Scales, Icon Sets, then New Rule, Clear Rules, Manage Rules.

Highlight Cells Rules

Numbers, text, dates, duplicates

Numbers

Greater Than, Less Than, Between, Equal To.

Text Contains

Matches part of a name. "Arif" highlights both Oneeb Arif and Anas Arif.

Exact Match Fix

Manage Rules > Edit Rule > type the full name "Oneeb Arif" > OK.

A Date Occurring

Yesterday, today, tomorrow, last 7 days... Type a date, then =A2+1, or use =DATE(2026,10,20).

Duplicates

Duplicate Values highlights repeats. Choose Unique to see single entries.

Top/Bottom Rules

Find the best and worst fast

`Top 10 Items`	**Highest values**
`Top 10 %`	**Highest percent**
`Bottom 10 Items`	**Lowest values**
`Bottom 10 %`	**Lowest percent**
`Above Average`	**Beats the average**
`Below Average`	**Under the average**

Change the 10 to any number, like Top 3 or Bottom 5.

Visual Rules

Data Bars, Color Scales, Icon Sets

Data Bars

A bar inside the cell. Bigger number = longer bar (20, 50, 80, 100).

Color Scales

Background color by value: low = red, medium = yellow, high = green.

Icon Sets

Arrows or symbols by value: 80+ marks = high, 50 = medium, 20 = low.

Data Bars show size, Color Scales show a heat map, Icon Sets show status.

Manage Your Rules

Bottom of the Conditional Formatting menu

New Rule

Build your own rule, including custom formulas.

Manage Rules

See, edit, reorder or delete every rule on the sheet.

Clear Rules

Remove rules from the selected cells or the entire sheet.

You can stack multiple conditions on one column, and change the format from the rule's Format button.

Custom Number Formats

Show Rs. and PKR in your numbers

`"PKR "#,##0.00`	PKR 1,250.50
`"RS. "#,##0`	RS. 1,250
`#,##0`	1,250
`0.00`	1250.50
`0%`	45%

Steps: select the column > Ctrl+1 > Number > Custom > type the code in Type > OK. Text goes inside quotes.

AI Add-ins in Excel

Use ChatGPT or Claude inside Excel

Step 1

File > Options > Customize Ribbon > Main Tabs > tick Developer > OK.

Step 2

Developer tab > Add-ins > Open Store. Search ChatGPT or Claude and click Add (a paid plan may be needed).

Step 3

The add-in shows on the Home tab. Sign in, select cells or the sheet, type a prompt and it applies.

Always double-check AI results before using them in a report.

Day 2
Done!

Formulas, functions, paste special and conditional formatting.

Which Excel function do you use the most?
Tell me in the comments.

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